STIVEN CATALYST

SECOND SERIES · NOVEMBER 2026

# MANAGEMENT REVIEW

Management without theatre.




STRATEGY

## SCENARIOS: A PLAN FOR MORE THAN ONE FUTURE

Why Wack called accurate forecasts dangerous, how use of the tool rose and fell, two uncertainties and four worlds, what scenarios changed in experts, how prepared firms fared, and a card for your own scenarios.

**1 in 3** of Shell's critical decision centres, at most, were acting on its 1972 scenarios a few months later. The rest found them interesting and carried on as before.

WACK, 1985

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# In this issue

Most plans rest on one forecast, and most of the time that is enough. Scenarios are for the times it is not: they let you plan for several futures at once without pretending to know which one will come.

Pierre Wack, who headed Shell's business environment division from 1971, found that good scenarios were not enough: they had to change how managers saw the world. In Bain's surveys, the tool's use rose to 70% in 2002 and fell to 19% by 2017. Paul Schoemaker shows how to cross two uncertainties into four worlds. In field studies by MIT researchers, experts changed most of their judgments but not their confidence, and leaned more towards flexible options. A longitudinal study links preparing for the future with better results.

*Stiven Janaqi* EDITOR · STIVEN CATALYST


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## HOW TO READ THIS ISSUE

**Figure**
Every figure has its source and year at the foot of its page.

**Our reading**
Where the editors interpret rather than the research, it says so.

**Practice**
The steps and the card are proposals to try, not research results.

COVER STORY

# Water on a stone

Pierre Wack headed the business environment division of Shell's planning department from 1971 to 1981. In 1985 he opened his account in Harvard Business Review with a warning about forecasts:

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> “Forecasts are not always wrong; more often than not, they can be reasonably accurate. And that is what makes them so dangerous.”

They fail, he wrote, when they are needed most: when the environment shifts. Shell's first scenarios only combined obvious uncertainties and gave managers nothing to act on. Those of 1972 caught top management's attention, yet months later no more than a third of the critical decision centres were acting on them. Unless they changed the picture of reality in managers' heads, scenarios were like water on a stone.

**01**
**First generation**
the obvious uncertainties, quantified: good for questions, not for decisions

**02**
**Predetermined**
events that have happened, whose consequences have not yet unfolded

**03**
**Microcosm**
the decision maker's mental model, the real target of a scenario

OUR READING

A scenario that nobody acts on is a report. It passes the test only when someone decides differently.

Source: Pierre Wack, Harvard Business Review 63(5), 1985

All of this is Wack's own account. Of 1973 the article says only that Shell was prepared for the eventuality, if not the timing, of the oil crisis.

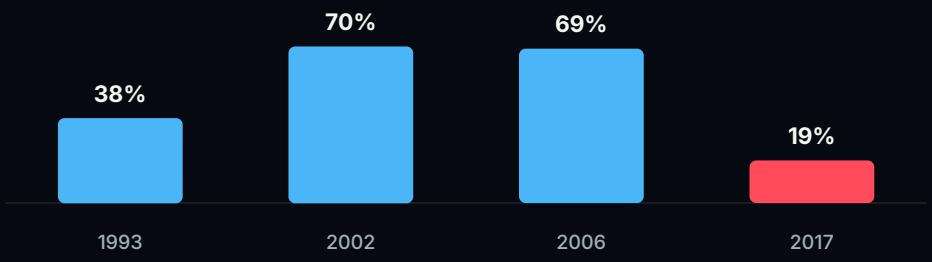
More in the essay High-volume days: the standard under pressure

THE NUMBERS

# In fashion after the shock

Since 1993 Bain & Company has asked executives around the world which management tools they use. Scenario and contingency planning counts as one tool. The share of respondents who used it:

EXECUTIVES USING SCENARIO AND CONTINGENCY PLANNING, BAIN SURVEYS




Bain links the rise after 2001 to the attacks of 11 September; in 2007 Darrell Rigby added Hurricane Katrina. In 2017 managers named it one of the two tools whose use would grow most. They had said the same in 2014, Bain notes, and neither tool then made the top ten.

OUR READING

A tool picked up after a shock and put away in calm years is serving as insurance rather than as a way of thinking.

Sources: Darrell Rigby, Bain & Company (first published on Forbes.com), 2007; Darrell Rigby & Barbara Bilodeau, Bain & Company, 2007; Darrell Rigby & Barbara Bilodeau, Bain & Company, 2018; Paul J.H. Schoemaker, Sloan Management Review 36(2), 1995

Self-reports, with a different sample each time (1,268 managers in 2017). The tool joins scenarios with contingency plans, which Schoemaker keeps apart: a contingency plan looks at one uncertainty, scenarios at several together.

THE MODEL

# Two uncertainties, four worlds

Paul Schoemaker of the Wharton School sets out ten steps, from the scope through trends and uncertainties to decision scenarios. One shortcut is to cross the two most important uncertainties, as Anglo American did in South Africa in 1984:

US AND USSR

| | |
|---|---|
| **Imperial Twilight**<br>Unrest in the Middle East and the spread of AIDS fitted this world better than the others.<br>ARMS RACE · TRADE ACCOMMODATION | **Dropped**<br>Judged implausible: the US would not take on both at once.<br>ARMS RACE · TRADE CONFLICT |
| **Industrial Renaissance**<br>A name captures the essence of a world and makes the story easy to remember.<br>DÉTENTE · TRADE ACCOMMODATION | **Protracted Transition**<br>For each world, Anglo estimated growth and which countries would succeed.<br>DÉTENTE · TRADE CONFLICT |

US AND JAPAN

## Schoemaker's four tests of a scenario set

1. relevant: it connects with the users' mental maps and concerns
2. internally consistent, and seen to be so
3. archetypal: generically different futures, not variations on one theme
4. each a state that can last for some time, not a passing moment

Source: Paul J.H. Schoemaker, Sloan Management Review 36(2), 1995

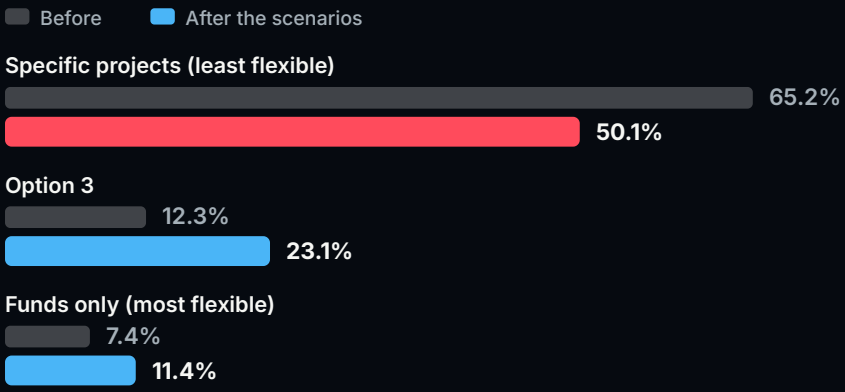
The steps, the tests and the Anglo American case are as Schoemaker reports them (1995); the short notes in the cells are the editors' summary.

WHAT THE RESEARCH SAYS

# Not more sure, more flexible

Shardul Phadnis and colleagues at MIT tested scenarios with field experts, from carrier owners to state planners, who judged real US transport investments 30 years ahead. In one workshop, 27 experts chose how flexibly to invest:

SHARE OF 351 RECOMMENDATIONS, BY OPTION




In another workshop, 71.3% of judgments changed after one scenario, yet after all of them average confidence was the same (0.811 and 0.814). Earlier tests had used students: Schoemaker's 68 MBA students widened their ranges by about half, and Meissner and Wulf found less framing bias among 252.

OUR READING

Scenarios made the experts neither surer nor less sure. They kept more doors open.

Sources: Shardul Phadnis, Chris Caplice, Yossi Sheffi & Mahender Singh, Strategic Management Journal 36(9), 2015; Paul J.H. Schoemaker, Sloan Management Review 36(2), 1995; Philip Meissner & Torsten Wulf, Technological Forecasting and Social Change 80(4), 2013 (via Abstract at IDEAS/RePEc; Phadnis et al., 2015)

No control group: the workshop was the treatment. The shifts were significant except for the most flexible option ($p = 0.056$).

HOW IT IS MEASURED

# Prepared or not

In 2008 René Rohrbeck and Menes Etingue Kum measured how prepared 83 large European firms were for the future: how much foresight their environment called for, against how mature their practices were. In 2015 they looked at the results.

83 EUROPEAN FIRMS, 2008




The vigilant firms averaged 16% profitability, against 12% for their industries. Their market value grew by 75% from 2008 to 2015, against 25% for the sample; the firms with more practice than they needed shrank by 6%.

HYPOTHETICAL EXAMPLE, A PARCEL DEPOT, ONE YEAR AHEAD

| | |
|---|---|
| **World** | "Busy and short-handed": volume up, drivers hard to find |
| **Signal** | applicants per opening, checked every month |
| **Trigger** | below 2 for two months: start training reserves |

Both the case and the figures are made up.

Sources: René Rohrbeck & Menes Etingue Kum, Technological Forecasting and Social Change 129, 2018; Paul J.H. Schoemaker, Sloan Management Review 36(2), 1995

Profitability for 70 firms, market value for 42 listed ones; foresight is wider than scenarios. The idea of one signal per scenario follows Schoemaker.

TOOL OF THE ISSUE

# The scenario card

One decision, one horizon. Write down what is already certain before what is not, give each world a name, and decide now which signal will tell you it is coming.

**01 Decision and horizon** what we must decide, and how far ahead we look

**02 What is already certain** trends everyone in the room accepts; the rest goes below

**03 Two uncertainties** the two that matter most, each with two plausible outcomes

**04 Four worlds** a name for each; strike one out only if it cannot happen

**05 Moves** useful in every world and wasteful in none; then those for one world only

**06 Signals and review** one early sign per world, who watches it, the date we look again

Sources: Pierre Wack, Harvard Business Review 63(5), 1985; Paul J.H. Schoemaker, Sloan Management Review 36(2), 1995; Shardul Phadnis, Chris Caplice, Yossi Sheffi & Mahender Singh, Strategic Management Journal 36(9), 2015

A practice the editors propose, drawing on Wack's predetermined elements, Schoemaker's two uncertainties and early indicators, and the robust investments of Phadnis et al. The control chart helps you tell a real signal from noise.

Open the tool: Sigma & Control Chart stivencatalyst.com/tools/sigma-control-chart/

SOURCES AND METHOD

# Every figure has a source.

The figures in this issue come from the sources below. The year shows how recent each one is.


**Scenarios: Uncharted Waters Ahead**
Pierre Wack, Harvard Business Review 63(5), 1985
https://hbr.org/1985/09/scenarios-uncharted-waters-ahead

**The next best thing to a crystal ball**
Darrell Rigby, Bain & Company (first published on Forbes.com), 2007
https://www.bain.com/insights/the-next-best-thing-to-a-crystal-ball/

**Management Tools and Trends 2007**
Darrell Rigby & Barbara Bilodeau, Bain & Company, 2007
https://www.bain.cn/pdfs/200906150356217586.pdf

**Management Tools & Trends (survey of 1,268 managers, fall 2017)**
Darrell Rigby & Barbara Bilodeau, Bain & Company, 2018
https://www.bain.com/insights/management-tools-and-trends-2017/

**Scenario Planning: A Tool for Strategic Thinking**
Paul J.H. Schoemaker, Sloan Management Review 36(2), 1995
https://sloanreview.mit.edu/article/scenario-planning-a-tool-for-strategic-thinking/

**Effect of scenario planning on field experts' judgment of long-range investment decisions**
Shardul Phadnis, Chris Caplice, Yossi Sheffi & Mahender Singh, Strategic Management Journal 36(9), 2015
https://doi.org/10.1002/smj.2293

**Cognitive benefits of scenario planning: Its impact on biases and decision quality**
Philip Meissner & Torsten Wulf, Technological Forecasting and Social Change 80(4), 2013 · via Abstract at IDEAS/RePEc; Phadnis et al., 2015
https://doi.org/10.1016/j.techfore.2012.09.011

**Corporate foresight and its impact on firm performance: A longitudinal analysis**
René Rohrbeck & Menes Etingue Kum, Technological Forecasting and Social Change 129, 2018
https://doi.org/10.1016/j.techfore.2017.12.013


## EDITORIAL METHOD

Each figure was checked for its year, its publisher and what exactly it measures. Where the publisher's page could not be opened, the figure was checked against independent summaries and is marked “via”. The editors' interpretation is marked “Our reading”. Figures that could not be confirmed are not in the issue.



# MANAGEMENT REVIEW

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Every issue, one management question, checked against the best research.

## ALL ISSUES

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Management Review · No. 64 · November 2026 · Stiven Catalyst